

杭州电子科技大学学生考试卷 (B) 卷

考试课程	通信电路与系统		考试日期	2023 年 月 日		成绩		
课程号	A040095s		教师号			任课教师姓名		林弥、陈瑾、李杭、 刘艳、高一强、张伟、 周涛、苏江涛、刘国 华、关阳阳、孙朋飞
题目	第一大题	第二大题	第三大题	第四大题	第五大题	第六大题	第七大题	
得分								

I. Judgment questions (3' each)

1. When the operation frequency is lower than the resonance frequency, a series LC resonant circuit shows capacitive.

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2. Television, AM/FM radio and mobile communications belong to millimeter wave (EHF) communications.

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3. Hartley circuit is easier to adjust frequency than Colpitts circuit.

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4. The frequency shift in an FM signal is related to the frequency of modulation signal.

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5. For Class C power amplifier, the conduction angle 2θ is less than $\pi/2$.

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6. Overtone crystal oscillator includes the fundamental frequency and its all-integer multiples.

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7. Compared with AM, FM signals are power and bandwidth saving.

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8. DSB transmitter saves more transmitting power than standard AM transmitter, because it has only half bandwidth of standard AM transmitter.

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9. Paralleling a resistor between two ends of LC resonant circuit will reduce the quality factor.

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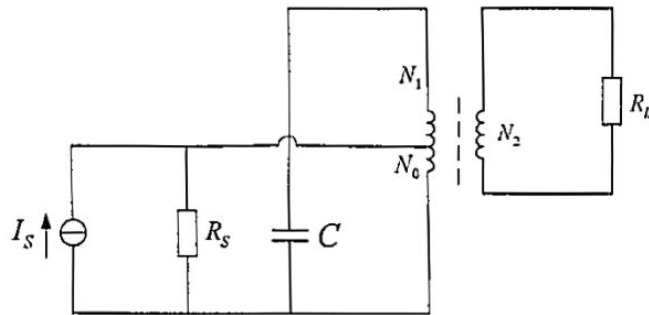
10. Angle modulation circuit is a linear spectrum shifting circuit.

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II. Calculation

1. (8') As shown in the figure, the resonant circuit $f_0 = 600 \text{ kHz}$, $Q_0 = 120$, the access factor in primary coils $n_1 = N_1/N_0 = 3$, the access factor between primary and secondary coils $n_2 = N_1/N_2 = 9$, $C = 180 \text{ pF}$, $R_S = 20 \text{ k}\Omega$ and $R_L = 1.5 \text{ k}\Omega$.

- (1) Calculate the inductor L and the resonant resistance R_0 (R_L and R_S are negligible).
- (2) Calculate the loaded quality factor Q_L and bandwidth $B_{0.7}$.

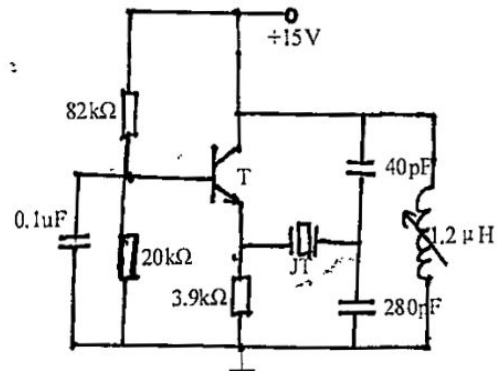


2. (8') A resonant power amplifier with operating at critical point state. $E_C = 18 \text{ V}$, current conduction angle $\theta = 70^\circ$, $\alpha_0(70^\circ) = 0.253$, $\alpha_1(70^\circ) = 0.436$, $U_{ces} = 3.3 \text{ V}$ and $I_{cmax} = 4.6 \text{ A}$.

- (1) Please calculate the DC power P_{DC} , power dissipation at the collector P_c , collector efficiency η_c and optimal load resistor R_{cp} .
- (2) To potentially prevent the transistor from burning out, what should we do for the equivalent load resistor?

3. (6') The oscillation circuit is shown below.

- (1) Draw its AC equivalent circuit
- (2) Is this crystal oscillator in parallel mode or series mode?
- (3) Explain the roles of crystal.
- (4) Calculate the feedback factor F of the circuit.



4. (15') The spectrum diagram of an AM signal is shown in Fig. (a)

- (1) Please give the time-domain expression of this AM wave.
- (2) Please calculate the whole power with unit resistor, and the bandwidth B .
- (3) Please give the advantage and disadvantage of this kind of AM.
- (4) Fig. (b) is the compositions diagram of AM receiving system. What do (a), (b) and (c) represent? If $f_L = 2465\text{kHz}$, calculate the down-mixing f_i .

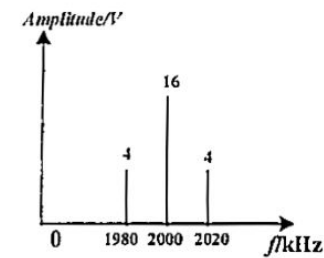


Fig. (a)

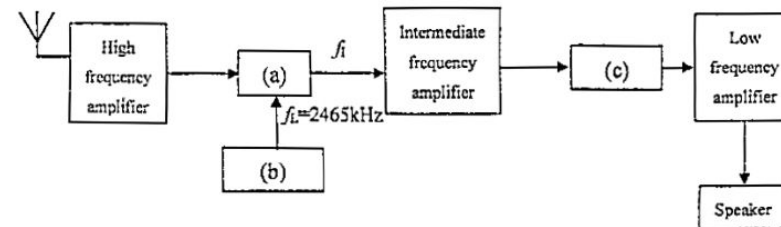


Fig. (b)

5. (16')

- (1) Please draw the basic circuit of the diode large signal peak envelope detector.
- (2) If $R_L = 8 \text{ k}\Omega$, $m_a = 0.4$, $f_c = 600 \text{ kHz}$ and the highest frequency of modulation signal $F = 280 \text{ Hz}$, how is the capacitor chosen to prevent the diagonal distortion?
- (3) If the detection efficiency $\eta_d = 0.97$, calculate the maximum equivalent input resistance R_{in} .
- (4) How is the input resistance of next-stage low-frequency amplifier chosen to prevent negative peak clipping distortion?
- (5) To demodulate the DSB and SSB signals, what kind of circuit can be selected? and draw the block diagram of its most commonly used circuit.

6. (14') The frequency of modulation signal is 300 Hz, its amplitude is 2.6 V and the angle modulation factor is 50.

- (1) For FM signals, write the frequency modulation factor m_f and calculate the bandwidth;
- (2) For PM signals, write the phase modulation factor m_p and calculate the bandwidth;
- (3) When the frequency of modulation signal is reduced to 200 Hz and its amplitude is increased to 3.8 V, calculate the updated modulation factor and bandwidth of the FM and PM signals, respectively;
- (4) Please draw the block diagram of slope frequency discriminator.

7. (3') Please list three interferences produced by the nonlinearity of mixer on communication environment.

B 卷答案

- I. 1, (✓) 2, (X) 3, (✓) 4, (X) 5, (X)
6, (X) 7, (X) 8, (X) 9, (✓) 10, (X)

II: 1. $L = \frac{1}{(2\pi f_0)^2 C} = \frac{1}{(2\pi \times 600 \times 10^3)^2 \times 180 \times 10^{-12}} = 391.3 \mu H$ ①'

$R_0 = Q_0 / 2\pi f_0 C = 176.93 k\Omega$ ①'

$R_\Sigma = R_0 // R'_S // R'_L$ ①'

$R'_S = n_1^2 R_S = 9 \times 20 \times 10^3 = 180 k\Omega$ ①'

$R'_L = n_2^2 R_L = 81 \times 1.5 \times 10^3 = 121.5 k\Omega$ ①'

$\therefore R_\Sigma = 51.45 k\Omega$ ①'

$Q_L = \frac{R_\Sigma}{\omega_0 L} = 29.4$ ①'

$B_{0.7} = \frac{f_0}{Q_L} = \frac{600 \times 10^3}{29.4} = 20.41 kHz$ ①'

2. 1) $P_{DC} = E_C I_{C0} = E_C \cdot I_{Cmax} \cdot \alpha_0(70^\circ) = 18 \times 4.6 \times 0.253 = 20.95 W$ ①'

$U_{cm} = E_C - U_{CES} = 18 - 3.3 = 14.7 V$ ①'

$P_0 = \frac{1}{2} U_{cm} \cdot I_{am} = \frac{1}{2} U_{cm} \cdot I_{Cmax} \cdot \alpha_1(70^\circ) = 14.74 W$ ①'

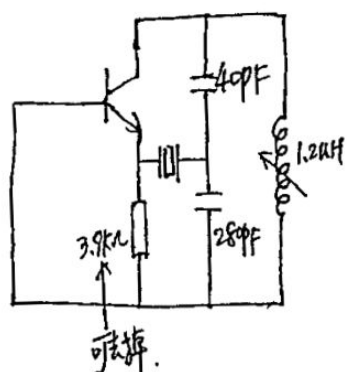
$\therefore P_C = P_{DC} - P_0 = 20.95 - 14.74 = 6.21 W$ ①'

$\eta_C = \frac{P_0}{P_{DC}} = 70.3\%$ ①'

$R_{cp} = \frac{U_{cm}}{2P_0} = 7.3 \Omega$ ①'

2) 我们应该避免等效阻抗太小。

3. (1)



(3)'

(2) 串联型晶体振荡器 ①'

(3) 晶体等效于短路元件 ①'

$$(4) F = \frac{U_f}{U_o} = \frac{1}{j\omega_0 280pF} / \left(\frac{1}{j\omega_0 40pF} + \frac{1}{j\omega_0 280pF} \right) = 0.12$$

$$4. 1) U_{AM}(t) = 20(1 + 0.5 \cos(2\pi \times 2 \times 10^4 t)) \cos(2\pi \times 2 \times 10^6 t) \text{ (V)}$$

$\uparrow \quad \quad \uparrow \quad \quad \uparrow \quad \quad \uparrow$
 $\text{①}' \quad \quad \text{①}' \quad \quad \text{①}' \quad \quad \text{①}'$

$$2) P_c = \frac{1}{2} \frac{U_{cm}^2}{R} = \frac{1}{2} \times 16^2 = 128 \text{ W} \quad \text{①}'$$

$$P_{side} = \frac{1}{2} m_a^2 P_c = \frac{1}{2} \times 0.25 \times 128 = 16 \text{ W} \quad \text{①}'$$

$$P_{总} = P_c + P_{side} = 144 \text{ W} \quad \text{①}'$$

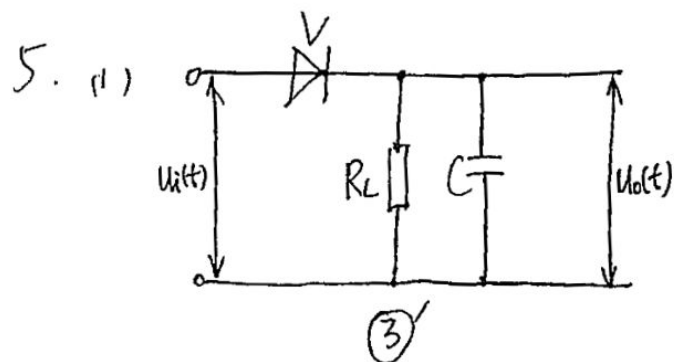
$$B = 2 \times 2 \times 10^4 = 40 \text{ kHz} \quad \text{①}'$$

4) (a) 混频器 ①' (b) 本地振荡器 ①' (c) 检波器 ①'

$$f_I = |f_L - f_c| = 2465 \text{ kHz} - 2000 \text{ kHz} = 465 \text{ kHz} \quad \text{②}'$$

3) 优点: 设备简单, 易于接收 ①'

缺点: 调制功率效率低 ①'

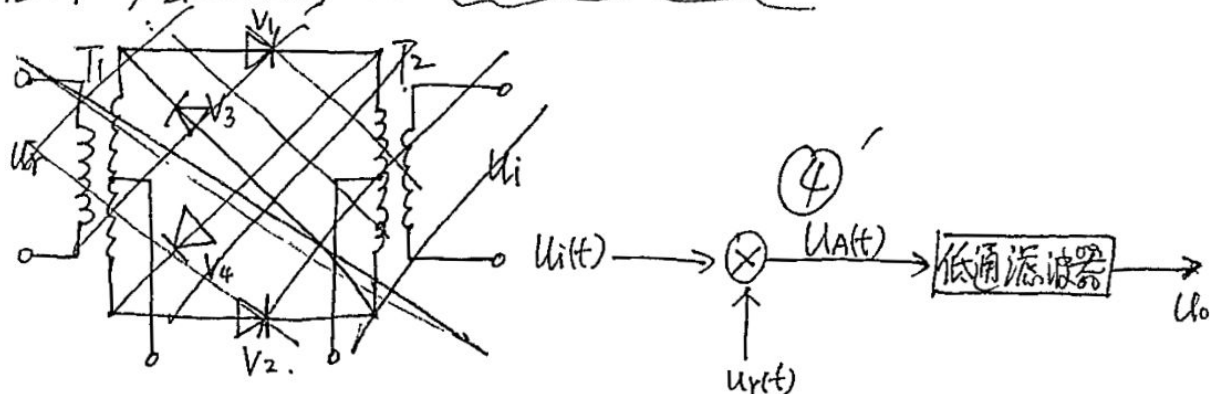


(2) $R_L C \gg \frac{1}{\omega_c} \Rightarrow C \gg \frac{1}{R_L \omega_c} = \frac{1}{8 \times 10^3 \times 2\pi}$
 $\approx 33.2 \text{ pF}$ (1)
 $C < \frac{1-m_a^2}{m_a} \times \frac{1}{F \cdot R_L \cdot 2\pi}$
 $= \frac{\sqrt{0.84}}{0.4 \times 2\pi} \times \frac{1}{280 \times 8 \times 10^3} = 0.163 \mu\text{F}$ (2)

(3) $R_{in} \approx \frac{R_L}{2\eta_d} = \frac{8 \times 10^3}{2 \times 0.97} = 4123.7 \Omega$ (2)'

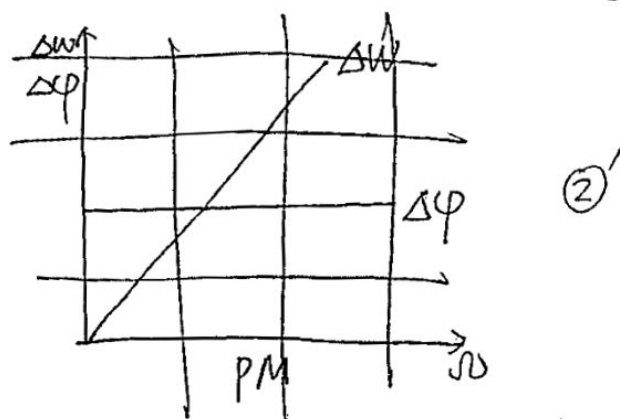
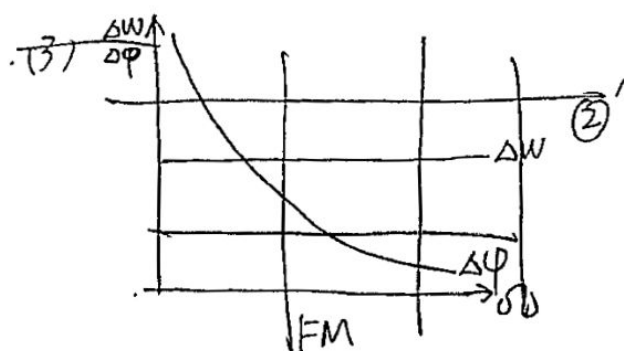
(4) $m_a \leq \frac{R_i}{R_i + R_L} \Rightarrow R_i \geq \frac{m_a R_L}{1-m_a} = \frac{0.4 \times 8 \times 10^3}{1-0.4} = 5.33 \text{ k}\Omega$ (2)'

(5) 二极管环形乘法器 ~~电路~~ 同步检波电路 (1)'



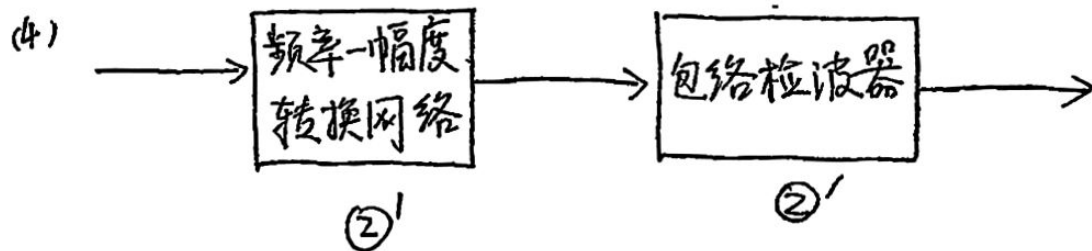
6. (3) $m_{f1} = \frac{k_f u_{r1}}{\omega_{f1}}, m_{f2} = \frac{k_f u_{r2}}{\omega_{f2}} \therefore \frac{m_{f1}}{m_{f2}} = \frac{200 \times 2.6}{300 \times 3.8} = 0.46 \therefore m_{f2} = 108.7$ (2)
 $B_{FM} = 2(m_{f2} + 1) \times 200 = 43.9 \text{ kHz}$ (1)'

~~且~~ $m_{p1} = k_p u_{r1}, m_{p2} = k_p u_{r2} \therefore \frac{m_{p1}}{m_{p2}} = \frac{2.6}{3.8} \therefore m_{p2} = 73.53$ (2)'
 $B_{PM} = 2(m_{p2} + 1) F = 29.8 \text{ kHz}$ (1)'



1) $m_{f1} = 50$ (1)' $B_{FM} = 2(m_{f1} + 1) \times 300 = 30.6 \text{ kHz}$ (1)'

(2) $m_{p1} = 50$ (1)' $B_{PM} = 2(m_{p1} + 1) \times 300 = 30.6 \text{ kHz}$ (1)' 3)



7. 1) 组合频率干扰 (干扰哨声)

2) 副波道干扰

3) 交叉调制干扰

4) 互相调制干扰

(3)' 列出3个即可